Greecon
SH.P.K

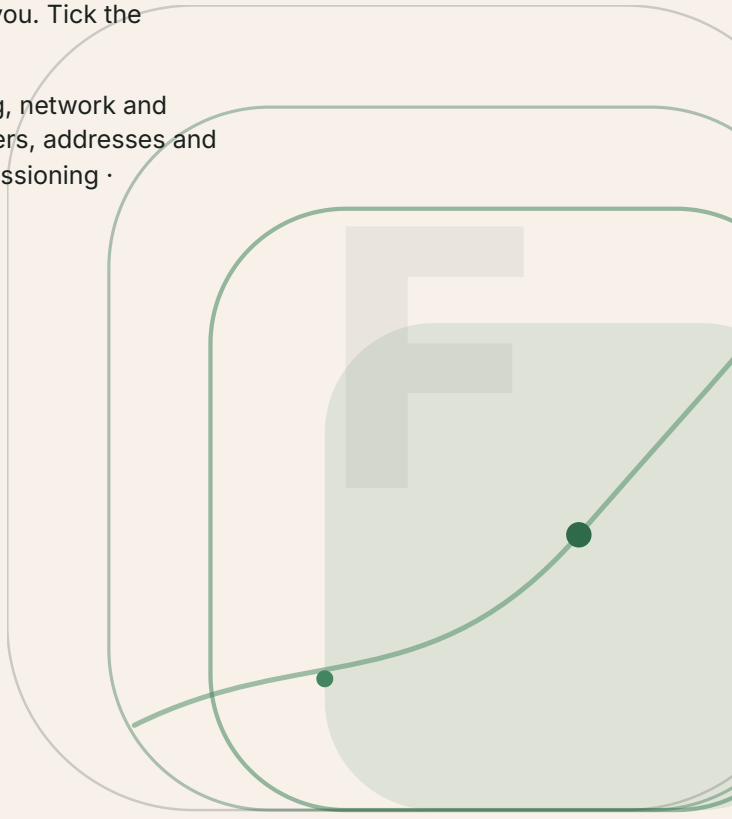
F

SITE PACK F

Farm or greenhouse with irrigation

Soil and weather, with the water supply behind it. Everything for one kind of site in one pack: what to buy, the survey, the diagrams, install day step by step, the platform steps, the commissioning tests and the handover sheet.

FOR	The project manager at this kind of site, the electrician, and the customer's owner at handover
PRINT	The whole pack, for the site in front of you. Tick the boxes as you go
COVERS	Bill of materials · survey · system, wiring, network and enclosure diagrams · install day · numbers, addresses and ranges · the twelve setup steps · commissioning · handover



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Greecon Sh.p.k · Durana Tech Park · info@greecon.earth

Contents

Site pack F Farm or greenhouse with irrigation 3
At a glance 3
How to read this pack 4
What it measures 5
What to buy and bring 5
The project manager brings 6
Survey 7
Ask, in this order 7
Look at, and photograph 7
Survey form 8
The drawings 9
At the office 13
In Greecon 13
The box on your desk 14
The numbers and the converter 14
Install day 16
Numbers, addresses and ranges 17
The numbers on the cable 17
Registers for the probe and the sensor: all of them from the makers' manuals 18
The pump contact module and the ranges 18
The platform's setup list 19
Step 1 Order form signed and data terms agreed 19
Step 2 Prices from the customer's bills 20
Step 3 Install the box 20
Step 4 Where each device is on the network 21
Step 5 Which register holds which reading 22
Step 6 Every device answers the box 22
Step 7 Readings arrive and look right 23
Step 8 Someone is told when the site needs attention 23
Step 10 Alert limits agreed with the customer 24
Step 11 Electrical work done by a licensed electrician 24
Step 12 Commissioned 24
Commissioning 25
Handover 27
Handover record 29
The three most likely problems 30

SITE PACK F

Farm or greenhouse with irrigation

Soil and weather, with the water supply behind it. A complete pack for one kind of site, in the order of the work: what to buy, the survey, the drawings, the office, install day step by step, the platform's setup list, the commissioning tests and the handover. Print all of it and tick the boxes as you go. The electrician does everything on the mains side; you do the low voltage and the software; Greecon only reads.

At a glance

WHO IT SUITS
A farm or a greenhouse with irrigation: soil moisture and weather at the field, with the water supply behind it (a tank, a pump, the pressure in the pipe).

WHAT IT MEASURES
The moisture of the soil (one probe for each zone), the air temperature and humidity, the tank, the line pressure and, if there is a pump to watch, the pump's flow and whether it runs.

PLANNING ESTIMATE
About EUR 700 to 1,500.

TIME
Survey about 1 hour, with the grower. Install day 4 to 6 hours. Commissioning about 1 hour. Handover 30 minutes.

TEMPLATE IN THE PLATFORM
Farm or greenhouse with irrigation

STANDARD SETTINGS
Soil probe 6, weather sensor 7, water module 3, pump contact module 5 (if there is a pump to watch), all set before the kit leaves the office. 9600 baud, 8 data bits, no parity, 1 stop bit. Converter on port 502, in Modbus TCP to RTU gateway mode. A cellular router is the likelier case in a field.

STOP

Greecon gives no irrigation advice until the grower's own "start irrigating below" value is entered. That number depends on the soil, the crop and the probe's depth, and nobody can set it for the grower. After the first week of readings, ask the grower what reading means "time to water", and enter it as the limit (step 10 of the setup list). **Frost works the same way:** ask at what air temperature frost damages the crop, and in which weeks; enter it as **Frost alert at or below** on the air temperature (step 10). It opens an Act now alert, so agree who is woken. Map the air temperature as a signed number (int16), or a frosty morning reads as a hot afternoon.

The order of the work: survey with the grower, buy, prepare at the office, install day, the platform's setup list, commissioning, handover. Then the morning check, and the day 1, 3 and 7 calls.

STOP

Six rules for the whole job.

1. **Greecon only reads.** It measures; it never switches anything. Never connect a wire from Greecon's equipment to a pump starter, an inverter's control, a valve or any safety circuit.
2. **Anything inside a distribution board is the electrician's work,** under the customer's lock-out procedure, with the project manager watching and writing it down. Nobody works on a live conductor.
3. **The project manager does the low-voltage and software side only:** the box, the network, the converter, the data cable (run to the board; the electrician connects its ends at the meters, because their terminals are inside the board), the labels, and the 12 or 24 V supply from a plug-in supply. The electrician wires the 4 to 20 mA loops, with their 24 V supply, as the module's manual draws it.
4. **Never touch the solar panels' wiring (DC), a tank you must climb into, wastewater or chemicals.** The customer's own staff or installer does that. If the only place for a sensor is there, say it is not possible and write why.
5. **Greecon is not a safety system.** The protection already in the customer's equipment (float switches, relief valves, trips) stays the safety layer.
6. **Write everything down at the time.** Photograph every meter's label and display, the wiring before the board is closed, the converter's settings and the box's label.

How to read this pack

Mark	What it means
ELECTRICIAN	The electrician. Everything inside a distribution board, every current clamp and voltage tap, the pump's run contact, and the 4 to 20 mA loops.
PROJECT MANAGER	The project manager: low voltage and software.
CUSTOMER	The customer: access, people, and starting or stopping their own equipment.
☐	A box to tick when it is done. Print the pack and tick as you go.
Do, **Check**, **Stop**, **Tip**, **Say**	The shaded boxes: do this; check this; stop and call the electrician; a useful habit; words to say to the customer.
Figure F-1 and so on	Drawings of this kit. Every drawing is marked with the kit it is for. Terminal numbers are never drawn: follow the maker's manual.

What it measures

The template **Farm or greenhouse with irrigation** makes four devices. What each gives:

Reading in Greecon	Comes from	Unit	Standard alert limits
Soil moisture	Soil probe: a soil moisture probe that speaks RS-485 Modbus, one for each zone, unit 6 (not an approved model: address and registers entered by hand)	%	None. **Start irrigating below** has no standard value: the grower's own is entered after the first week
Air temperature and **Humidity**	Weather sensor: a temperature and humidity sensor that speaks RS-485 Modbus, unit 7 (not an approved model: address and registers entered by hand)	degrees Celsius, %	None
Tank level	Water I/O module, unit 3: the level transmitter's 4 to 20 mA	%	Critical below 15 % (Act now); Low below 35 % (Check today)
Line pressure	Water I/O module, unit 3: the pressure transmitter's 4 to 20 mA	bar	None
Pump flow (if there is a pump to watch)	Water I/O module, unit 3: the flow meter's 4 to 20 mA	litres a minute	No flow at or below 0.2 litres a minute (Act now, once it has held for 30 seconds, and only when the box reads both a flow and the pump's running state)
Pump running (if there is a pump to watch)	Pump contact module (optional), unit 5: the pump starter's spare contact	on or off	None

A sensor whose maker does not document its registers cannot be mapped, so it cannot be used. Soil readings are only as good as the contact between the probe and the soil: the handover records the probe's depth and where it is.

What to buy and bring

	Item	Specification	Qty	Planning estimate (EUR)
☐	**Box**	Fanless x86 mini PC (see the box specification), 8 GB, 128 GB or more SSD or eMMC, 12 to 24 V DC input	1	180 to 300
☐	**Converter**	RS-485 to Ethernet, DIN rail, 9 to 36 V DC, Modbus TCP to RTU gateway mode	1	30 to 60 (basic) or 250 and up (industrial)
☐	**4 to 20 mA input module**	Waveshare Modbus RTU Analog Input 8CH (current version, the one that reads current), 8 inputs, 4 to 20 mA, 7 to 36 V DC	1	30 to 60

	Item	Specification	Qty	Planning estimate (EUR)
☐	**Level transmitter**	Submersible, vented cable, 4 to 20 mA, range at least the tank's height, wetted parts approved for drinking water (or a non-contact ultrasonic sensor above the tank)	1	120 to 400
☐	**Pressure transmitter**	4 to 20 mA, range to suit the pipe (0 to 10 bar is usual), stainless wetted parts	1	40 to 150
☐	**24 V DIN power supply**	30 to 60 W	1	25 to 50
☐	**Enclosure and fittings**	IP54 (IP65 outdoors), DIN rail, terminals, fuses or small breakers for the meters' voltage taps, glands	1	80 to 150
☐	**Cable and labels**	Shielded twisted pair for the RS-485 line, network patch cords, ferrules, labels	1 lot	40 to 80
☐	**Cellular router**	4G, DIN rail, data SIM. *The likelier case in a field.*	0 or 1	150 to 200
☐	**Soil moisture probe**	RS-485 Modbus, with a published register table, one for each zone	1 for each zone	40 to 120 each
☐	**Temperature and humidity sensor**	RS-485 Modbus, with a published register table	1	20 to 80
☐	**Pump contact module** *(optional)*	Waveshare Modbus RTU IO 8CH: eight isolated digital inputs, a dry contact is enough. Relay outputs never used. *If there is a pump to watch.*	0 or 1	30 to 60
☐	**Flow meter** *(optional)*	4 to 20 mA output. Needed for "the pump is running but no water is flowing". *If there is a pump to watch.*	0 or 1	150 to 500

About EUR 700 to 1,500. One soil probe for each zone. Ask three suppliers for the quote on the specification, not the brand, and keep the quotes. Buy and bench-test one kit before buying five. The reasons for each part, its settings and the mistakes to avoid are in the Hardware Reference (PDF 03).

The project manager brings

- ☐ A laptop
- ☐ A USB-to-RS-485 adapter (to number a module or a probe, and to talk to one device alone)
- ☐ A loop calibrator (or a 4 to 20 mA source)
- ☐ A multimeter
- ☐ Screwdrivers (slotted and Phillips, small and medium)
- ☐ Wire strippers and a ferrule crimper
- ☐ A network cable tester
- ☐ Cable ties and labels
- ☐ A head torch
- ☐ Spare fuses and terminals
- ☐ A phone for photographs

You do not need a tool for work inside a live board: that is the electrician's. The USB-to-RS-485 adapter is how you set a probe's or a sensor's unit number before it is buried or mounted. The spares kit (one for each project manager) goes in the car too.

Survey

About one hour, with the grower and, if the plan needs one, the electrician. Bring this pack, a phone for photographs, a tape measure, a torch and a notebook. No tools are needed to survey, and nothing is opened.

Ask, in this order

The answers decide what is measured and what it costs. Write them on the survey form that follows.

1. Who decides, and who gets called at night? Names, mobile numbers, emails: these become **Who is told**. Each alert message carries an *I have seen this* link, so nobody needs a login to acknowledge; the people who should also look at the screens get a login of their own, and a login needs an email address.
2. What went wrong in the last twelve months that cost money or a crop? Write it in their words. It is the reason they buy.
3. **Which crop, which soil, how deep are the roots, and where in the field is typical** (not the wettest or the driest spot)? The irrigation limit depends on them.
4. Where are the water tank, the pump and its starter? How tall is the tank? What is the pipe's pressure? Is there a pump to watch, and has its starter a spare "running" contact?
5. Is there a network socket in the pump house, or cellular signal? (Look at your phone there.)
6. Is there a power socket for the box, on a circuit that stays on?
7. How far is the probe's place from the box? The data cable is planned at under 100 metres. If it is further, a probe on a radio gateway that speaks Modbus is the alternative; the gateway's maker must publish its register table. Ask the founder before promising it.
8. Who is the customer's electrician, and can they be there on install day?

Look at, and photograph

- The field: where each probe goes, and the weather sensor's place.
- The tank, the pump, the starter, and any existing level or pressure gauge and flow meter.
- The pump house: where the box and the cellular router would sit, dry, 30 cm free, no heat, within reach of a socket.
- The route for the data cable from the field to the box (note any crossing of power cable).

CHECK

Go if there is a way to the internet (cable or cellular), a socket that stays on, a safe place for each meter, an electrician for the board work, and a person who answers when called.

No-go, politely, and write why, if the only place for a sensor is unsafe to reach, the customer wants something switched or controlled, or there is no responsible person to receive alerts.

☐ Go ☐ No-go, because: _____

Survey form

KIT F · FARM OR GREENHOUSE WITH IRRIGATION

Field	Answer
Customer, site name and address, date	
Contact: name, mobile, email (first, second, third to be told)	
What went wrong last year (their words)	
Crop; soil; root depth (cm)	
Zones: how many probes; where each goes (typical, not wettest or driest)	
Distance from the probe places to the box (m); wired or radio gateway	
Tank: height (m), water source; pump (kW); starter location; spare running contact on the starter; pipe pressure (bar)	☐ No pump to watch
Sensors needed: level, pressure, flow, pump running (and where)	
Network: customer's cable or cellular; signal in the pump house	
Power for the box: socket location	
Electrician: name, mobile, day	
Photographs taken (list)	
Go / no-go, and why	☐ Go ☐ No-go

The drawings

Every drawing is for kit F and says so. They show how the parts connect, not terminal numbers: follow the maker's manual for terminal numbers.

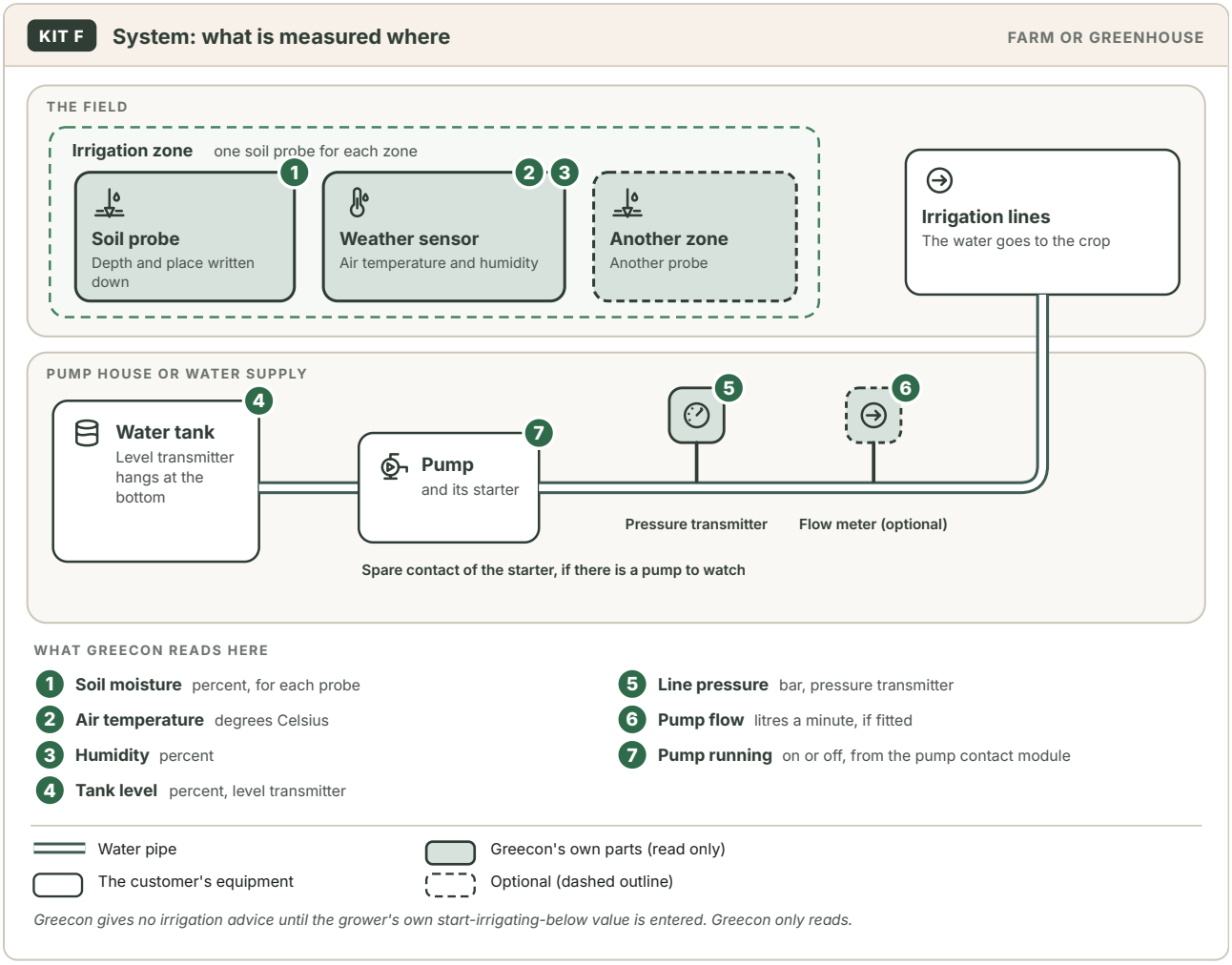
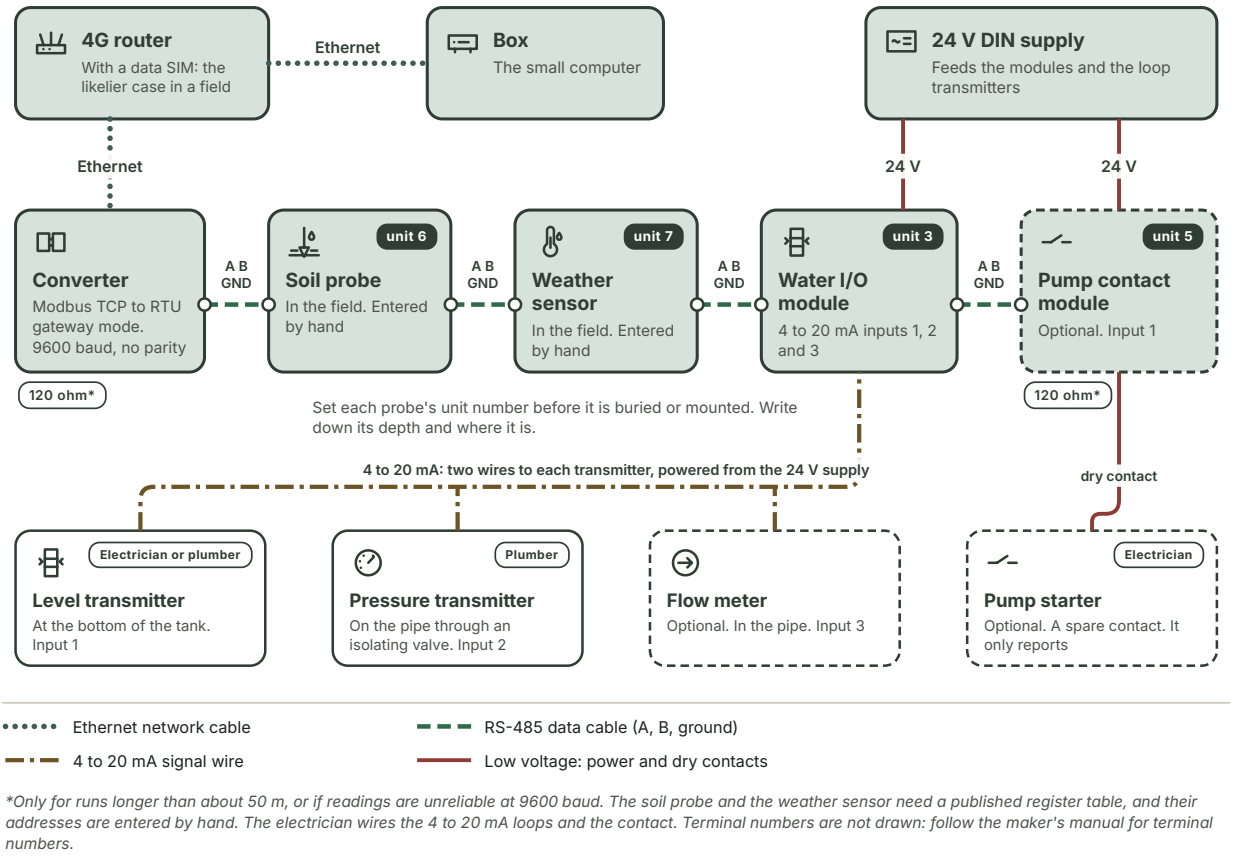


Figure F-1. Kit F, farm or greenhouse with irrigation. System: what is measured where. The dashed parts are optional.

KIT F **Wiring: the data cable and the signal wires** BUS AND SIGNALS



Only for runs longer than about 50 m, or if readings are unreliable at 9600 baud. The soil probe and the weather sensor need a published register table, and their addresses are entered by hand. The electrician wires the 4 to 20 mA loops and the contact. Terminal numbers are not drawn: follow the maker's manual for terminal numbers.

Figure F-2. Kit F. Wiring: one RS-485 cable from the converter through the soil probe, the weather sensor and the water module to the pump contact module, with the 4 to 20 mA loops into the water module and the pump starter's contact into the pump contact module.

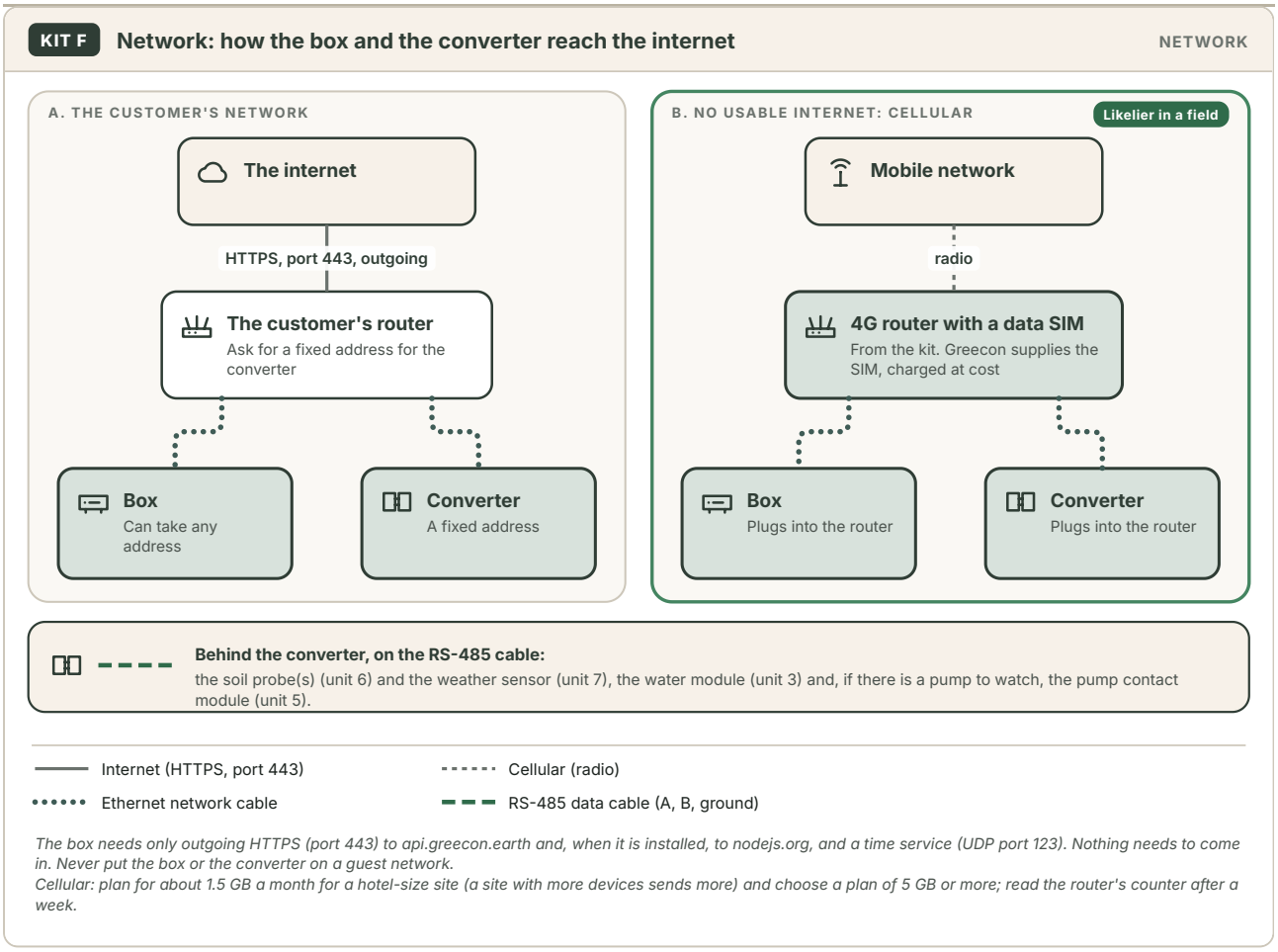


Figure F-3. Kit F. Network: a 4G router with a data SIM is the likelier case in a field; the customer's router is better where there is one.

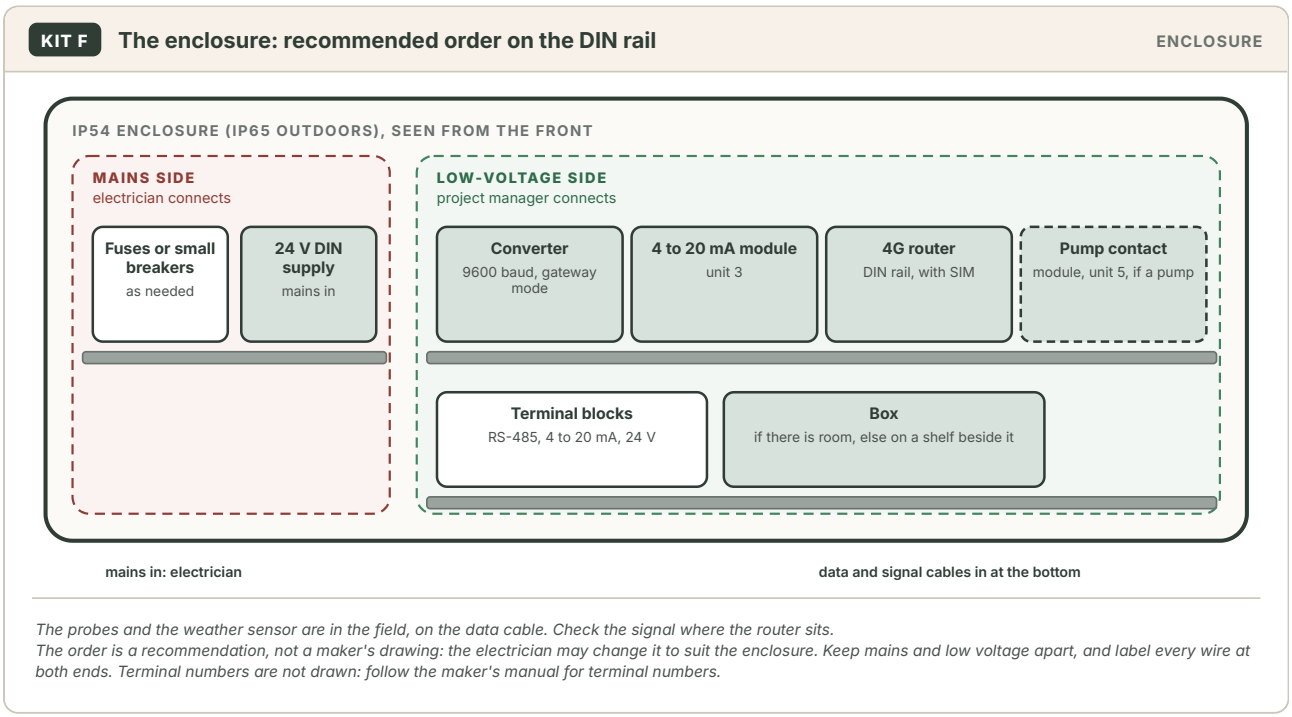


Figure F-4. Kit F. The enclosure: the recommended order of the parts on the DIN rail, mains side on the left.

The order on the rail is a recommendation, not a maker's drawing: the electrician may change it to suit the enclosure. Keep mains and low voltage apart, and label every wire at both ends.

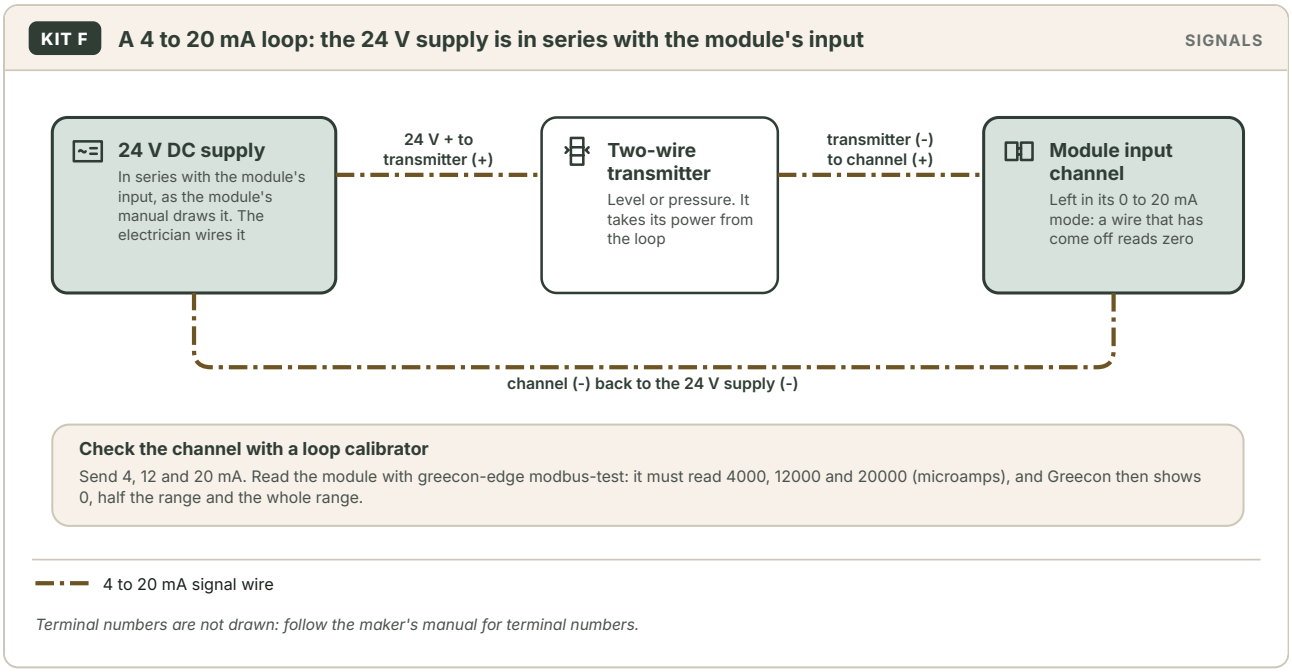


Figure F-5. Kit F. One 4 to 20 mA loop; the level, the pressure and the flow transmitter are each wired this way.

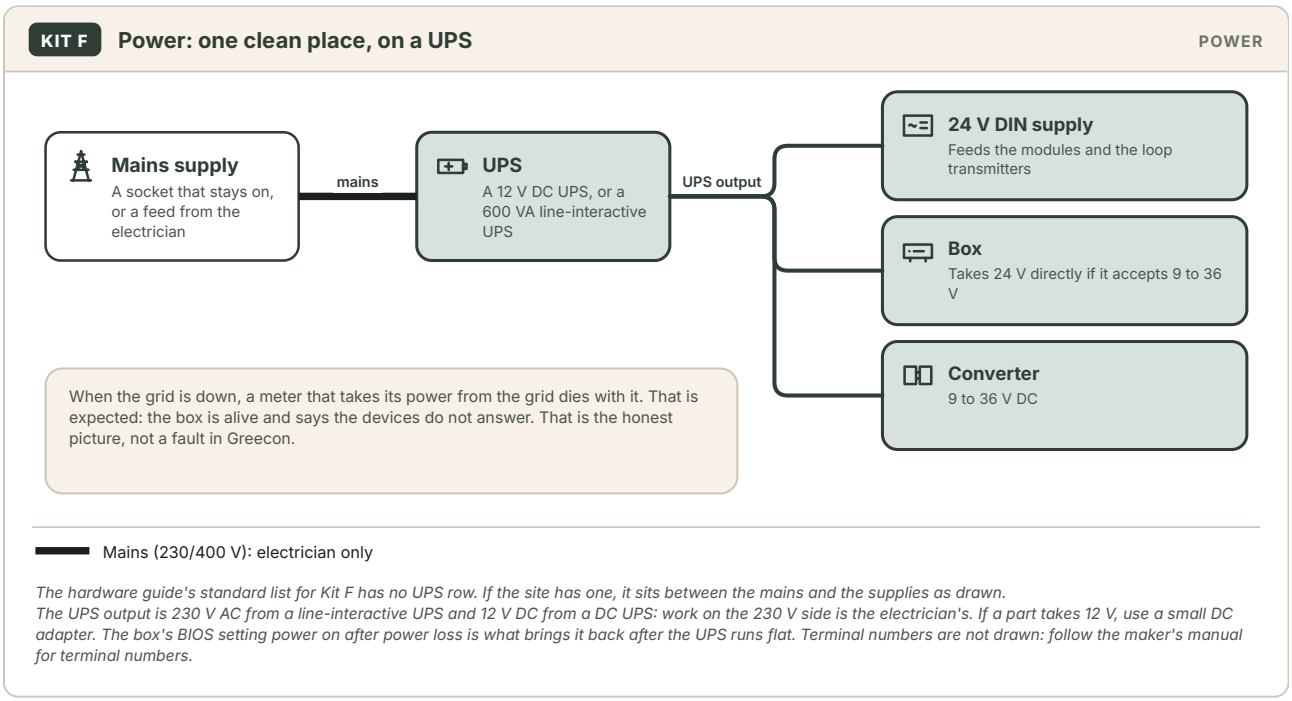


Figure F-6. Kit F. Power. The hardware guide's standard list for this kit has no UPS: decide at the survey and write it down.

At the office

Everything in this chapter is done at the office, before the site visit, and it makes install day shorter. About 30 minutes in Greecon and about two hours for the box, and more for the first kit's numbers and its desk test: time it, and write it down. PROJECT MANAGER

In Greecon

- [ ] **Sign in** at `https://app.greecon.earth/login` with **Greecon team**. You land on **Today**.

- [ ] **A new customer:** if they asked through the web page, their request waits under **Requests waiting** on **Today** (and on **Requests**): press **Onboard**. Otherwise open **Clients** and press **Onboard a client** (top right): company name, owner's name, owner's email, then **Onboard client**. A box appears, **Client onboarded — save this password now**: it is the owner's temporary password, shown once. Write it on the handover page or tell it to the owner by phone. Do not send it by email. If a message says someone already has an account with that email, the person is already a customer: add the new site to the customer they already have.

- [ ] **Open the customer for setup:** in the customer's row on **Clients** press **Open for setup** (on **Today** the same button is **Open account** for a site that needs a look and **Continue setup** for a site being set up). You land on the customer's **Sites** page, with a banner saying you are there as Greecon staff, setting it up. Everything you do is recorded under your own name. The setup session lasts four hours: after that, sign in again with **Greecon team** and press **Open for setup** again. What you saved is still there.

- [ ] **Make the site:** on **Sites** press **Add a site**. In the form (*New site from a template*) choose **What kind of site**: **Farm or greenhouse with irrigation**. The page lists what the template will contain. Enter **Site name** and **Where it is**, check **Currency of the bills** (it starts as ALL) and **The site's own clock** (Europe/Tirane), and press **Create the site and start setting it up**.

- [ ] **Write down that the papers are signed:** you land on the setup page, titled **Set up** and the site's name, which says `0 of N steps done`, and its first step is **Order form signed and data terms agreed**. With the signed order form and the agreed data terms in front of you, write in the box who signed for the customer and on what date (it cannot be left empty), and press **The papers are signed**. Nothing goes into a customer's building without both papers.

- [ ] **Print the install card:** the link **The install card** is at the top of the setup page. It is your list for the visit: what to bring, what each device is set to and its number, how it is wired, the converter's settings, and what the customer's network is asked for. At the bottom are three messages to copy: send the first to the customer's contact two or three days before, and the second to whoever looks after their network. Each is in English with an *Albanian draft* folded under it, and someone who writes Albanian every day reads the draft before it is sent.

- [ ] **Get the install line:** on the same page press **Add the box and get its install command**. A line appears, beginning `curl -fsSL https://api...` and ending in a code like `GRC-XXXX-XXXX-XXXX`. Copy it with the copy button. It is shown once. The code works once, for seven days. Press **I have copied it**.

STOP

The install line is the box's key. Anyone with the line can install a box as this site until it is used or has expired. Treat it like a password: not in email or chat, and pasted straight into the box. If it is lost, or the seven days have passed, press **Install a box, or replace this one** on the same step; it asks you to confirm and shows a new line. The old line then stops working, and the box that uses the new line takes the place of the box now installed.

The box on your desk

The operating system and the pasted install line are still a terminal job; everything after them is done on the setup page.

- ☐ **Install the operating system** on the box: Debian 12, Ubuntu 22.04 or later, or Raspberry Pi OS 64-bit, with the SSH server on, one user with a long password, and the time zone set to Europe/Tirane. The twenty-minute recipe and the BIOS settings are in the Hardware Reference (PDF 03), section 4.3.
- ☐ **Connect the box** to your office network and open a terminal on it (a keyboard and screen, or SSH).
- ☐ **Paste the install line exactly.** It runs in seven numbered steps and tells you each one: checking the box (and that it can reach Greecon and agrees with it about the time), installing Node.js, making the places the software uses, fetching and checking the Greecon software, giving the box its own credential, starting the service, and waiting to hear that Greecon can see the box. It finishes with `Done. Greecon can see this box.`
- ☐ **Refresh the setup page. Install the box** now says **Done**, with the box's name, its version and "Heard from N seconds ago". You did not press anything to make it so: the platform saw the box.
- ☐ **Label the box** with the site name and the date, and pack it. The box keeps its credential when it is moved: unplug it, carry it to the site, plug it in, and it reports from the new place without anything being typed again.
- ☐ **Pack the rest** of the kit from the shopping list, the tools, the spares kit, the printed pack and the printed install card.

The numbers and the converter

The kit is made of approved models, and every device has the same number on the cable in every kit. You set them here, at the office; on site you only check the labels and do not change them. The numbers for this kit are in the table under **Numbers, addresses and ranges** and on the install card. Everything on one cable is **9600 baud, 8 data bits, no parity, 1 stop bit**.

- ☐ **Set each meter's number on its own display,** one meter at a time (delivered as unit 1, two on one cable would answer over each other). The SDM630 is delivered at 9600 baud: check it. The SDM230 is delivered at a speed that depends on its version (the V1.6 manual says 2400): set it to 9600, on the Baud Rate page. Powering a meter on a bench is mains work: the electrician makes up the office's bench lead, and you never wire mains.
- ☐ **Number the two Waveshare modules** with the maker's setting tool and a USB-to-RS-485 adapter, one at a time (each is delivered as address 1, and there is no switch): **3** on the 4 to 20 mA module (Modbus RTU Analog Input 8CH) and **5** on the pump contact module (Modbus RTU IO 8CH). Use the version of the 4 to 20 mA module that reads current, not the one that reads voltage, and leave its inputs in their default 0 to 20 mA mode, on purpose: a 4 to 20 mA transmitter then reads 4000 to 20000 microamps, and a wire that has come off reads 0, below the 4 mA of an empty tank, which Greecon can see and say.
- ☐ **Set the converter** in its web page: mode **Modbus TCP to RTU (gateway)** (not transparent, not raw TCP, not Modbus RTU over TCP), 9600 baud, 8 data bits, no parity, 1 stop bit, TCP port 502. Its address on the customer's network is arranged on site: a fixed one is best, and Greecon finds the converter either way.
- ☐ **Label every device with its number** and the converter with its settings, and photograph the labels.

☐ **Test the kit on the desk.** The registers of the two Waveshare modules are from the maker's published tables and have not been tried on real hardware, so the first real kit is rehearsed on a desk before it is trusted (the Project Manager Field Manual (PDF 02), section 3.11). Anything that differs from this book is reported to Greecon.

TIP

If the line stops with `STOP:` it says what to check and how. If it stopped before step 5, *Giving this box its credential*, fix the cause and run the same line again: the code has not been used yet. From step 5 on, the code is used up, and the same line would stop with "Greecon did not give the software to this box". Then run the line again with no code at the end: `curl -fsSL https://api.greecon.earth/edge/install.sh | sudo bash -s --`. That repairs the box and keeps its credential. If it answers "This box has no code and has not been installed before", the box never got its credential: press **Install a box, or replace this one** and use the new line. The commonest stops are a box that cannot reach Greecon (no internet, a firewall that blocks outgoing HTTPS on port 443, or a clock so far out that secure connections fail), and a code already used or older than seven days. A clock only minutes out does not stop the install: it prints a warning, and the setup page later says its clock is N minutes fast.

TIP

The first time, rehearse the whole job on a desk with a laptop standing in for the kit: the Project Manager Field Manual (PDF 02), section 3.11. It takes an afternoon and finds the mistakes in the room instead of at the customer's.

Before the kit leaves the office, set the unit number of each soil probe (**6**) and of the weather sensor (**7**), with the USB-to-RS-485 adapter and the maker's instructions, and write them down: **set them before the probe is buried or the sensor is mounted.** Make every one different from the others and from the water module (3) and the pump contact module (5): a second probe, for a second zone, takes the next free number (8, then 9), written down.

Install day

About four to six hours. The electrician and the project manager work at the same time, in this order, so that nobody waits. The left edge of each row is black for the electrician, green for the project manager and sand for the customer.

	#	Who	Do this
☐	1	ELECTRICIAN · PROJECT MANAGER · CUSTOMER	**Before anyone touches anything:** the signed order form and the data terms are in your hand, the electrician is there, and the customer's lock-out is agreed. **Walk the site together,** with the grower. Agree where each probe and the weather sensor go (typical of the zone, not the wettest or driest spot), and where the enclosure sits. Photograph everything before you start.
☐	2	CUSTOMER	Show the way to the pump house, the tank and the starter. Have a socket that stays on ready.
☐	3	ELECTRICIAN	**Apply the customer's lock-out** on the circuit that feeds the pump house.
☐	4	ELECTRICIAN	**Connect the mains to the 24 V DIN supply** in the enclosure, as its manual shows, with a fuse or small breaker as needed.
☐	5	ELECTRICIAN	**If there is a pump to watch:** connect the pump's run contact (a spare auxiliary contact of the starter, as a dry contact) to **input 1 of the pump contact module**, as the module's manual draws it (its DI COM left unconnected). Never to a control circuit, and never to the module's relay outputs: the contact only reports.
☐	6	ELECTRICIAN · PROJECT MANAGER	**Fit the level and pressure transmitters** (and the flow meter, if there is one) with the customer's plumber for the wet parts, and **wire their 4 to 20 mA loops** with the 24 V supply in series, as the module's manual draws it: **level on input 1, pressure on input 2, flow meter on input 3.** The level transmitter hangs at the bottom of the tank and must be rated for drinking water. The pressure transmitter goes on the pipe through an isolating valve. Figure F-5. You check each is on its own input and number the wires at both ends.
☐	7	PROJECT MANAGER	**Mount the enclosure** and in it the converter, the water module, the pump contact module (if there is a pump to watch), the 24 V supply, the 4G router (and the box, if there is room, otherwise on a shelf beside it). Figure F-4.
☐	8	PROJECT MANAGER · CUSTOMER	**Place the soil probe(s)** at the depth and in the place agreed with the grower, in good contact with the soil, as the maker says; mount the weather sensor. **Write down each probe's depth and where it is, and photograph it.**
☐	9	PROJECT MANAGER	**Run the RS-485 data cable** from the converter to the soil probe(s) and the weather sensor, and on to the water module and the pump contact module: one cable, a daisy chain, no branches. Shielded twisted pair, the shield joined to ground at one end only, away from power cables. Figure F-2.
☐	10	PROJECT MANAGER	**Check the labels:** the number on each device against the table under *Numbers, addresses and ranges* (soil probe 6, weather sensor 7, water module 3, pump contact module 5). They were set at the office: do not change them. Photograph each label.

	#	Who	Do this
☐	11	PROJECT MANAGER	**Check the converter's label** (gateway mode, 9600 baud, 8 data bits, no parity, 1 stop bit, port 502) and **connect the network:** put the SIM in the 4G router, check the signal where it sits, and plug the box and the converter into its sockets (or into the customer's router if the pump house has one). Ask for a fixed address for the converter if it is on the customer's router, and write it down. Figure F-3.
☐	12	ELECTRICIAN PROJECT MANAGER	**Photograph before closing:** the wiring, the converter's settings, each device's label. Then close the enclosure.
☐	13	ELECTRICIAN	**Remove the lock-out** only when you both agree the work is closed up.
☐	14	PROJECT MANAGER	**Power up** the 24 V supply, the converter, the router and the box.
☐	15	ELECTRICIAN	**Stay until commissioning is done.**
☐	16	PROJECT MANAGER	**Open the customer for setup again** on your laptop (the four hours are over by now) and go down the platform's setup list: **Find equipment**, the probe and the sensor by hand, the ranges, and the electrician's name and licence number.

Numbers, addresses and ranges

The numbers on the cable were set at the office. The box finds the water module and the pump contact module on the network. It cannot recognise a soil probe or a weather sensor, so those two are entered by hand, with their registers, from the makers' manuals. For the transmitters you give the range printed on each label.

The numbers on the cable

All the devices are behind one converter: they share its address and port 502, and differ only by number. Check each label against this table on install day. **Do not change a number on site.**

Device in Greecon	Make and model	Number	Label checked	Address (the converter's)
Soil probe (zone 1)		6	☐	
Soil probe (zone 2, if any)		8 (the next free number)	☐	
Weather sensor		7	☐	
Water I/O module	Waveshare Modbus RTU Analog Input 8CH	3	☐	
Pump contact module (optional)	Waveshare Modbus RTU IO 8CH	5	☐	

For the soil probes and the weather sensor, under each device open **Enter the address by hand** and fill in its **IP address** (the converter's: the water module's line in the same step shows it, as *Reached at 192.168.1.50:502, unit 3*), its **Port** (502) and its **Unit number**.

Registers for the probe and the sensor: all of them from the makers' manuals

There is no ready-made profile for a probe or a sensor: each reading is mapped by hand from the maker's register table (step 5 of the setup list). Under each reading's name press **Choose the register**. The register kind is **Input** for a manual's 3xxxx and **Holding** for 4xxxx; the address is counted from zero (30053 is address 52). Write what you enter:

Reading	Register kind	Address	Kind of number	How it becomes the reading
Soil moisture (zone 1)				
Air temperature				
Humidity				

The pump contact module and the ranges

If there is a pump to watch, the pump contact module needs nothing typed: the template fills in its reading (digital input 1). Check it by eye: with the pump running, **Pump running** reads 1; stopped, 0. Test it from the box: `greecon-edge modbus-test --host 192.168.1.50 --unit 5 --type discrete --address 0`. With no pump to watch, press **The site does not have this: remove it** under it in step 4.

For the water module, one form for each transmitter (step 5 of the setup list), in the label's own words. Read the range from the label on the transmitter and measure how tall the tank is inside, from the bottom to the top of the water when it is full.

Reading	Input	The form asks for	Range on the label	Tank height
Tank level	1	Level transmitter: the range on the label (m or bar) and how tall the tank is inside (m)		
Line pressure	2	Pressure transmitter: the top of the range (bar)		
Pump flow	3	Flow meter: the top of the range (lpm, litres a minute)		

Greecon works out the register and the scale: 4 mA is 0 and 20 mA is the range (for the tank, range ÷ tank height × 100: a 0 to 5 m transmitter in a tank 4 m tall gives 125, and a full tank, at 16.8 mA, reads 100). **A transmitter shorter than the tank is refused:** it cannot see the top of the water. Tick **Not fitted at this site** for the readings the site does not have (no flow meter): the reading is taken away and nothing waits for it.

The platform's setup list

The platform keeps its own list for each site. Open the site's setup page (**Set up** and the site's name). It shows up to twelve steps, always in this order; a step that does not apply to the site is left out. Each step is **Done**, **To do**, **Waiting** for another step, or **Needs attention**, and the banner at the top, **Do next**, names the step to do next. The platform works most steps out by itself: you press buttons only for what a person must check or say. Near the top are links to the site page, to **The install card** (the card for the visit) and to the **Handover sheet**.

	Step	Title on the setup list	Done by
☐	1	**Order form signed and data terms agreed**	Project manager
☐	2	**Prices from the customer's bills**	Project manager (optional)
☐	3	**Install the box**	Project manager
☐	4	**Where each device is on the network**	Project manager
☐	5	**Which register holds which reading**	Project manager
☐	6	**Every device answers the box**	The platform, by itself
☐	7	**Readings arrive and look right**	The platform, by itself
☐	8	**Someone is told when the site needs attention**	Project manager
☐	10	**Alert limits agreed with the customer**	Project manager
☐	11	**Electrical work done by a licensed electrician**	Project manager, with the electrician
☐	12	**Commissioned**	Project manager

Below, each step says what to do, what you should see, and what to do if you do not.

Not on this kit. The platform leaves out a step that does not apply to the site, so the list for kit F has no step 9 (Grid meter direction checked). The steps keep their order; the count on the setup page is out of the steps that do apply, not out of 12.

Step 1 Order form signed and data terms agreed

PROJECT MANAGER

DO	Nothing goes into a customer's building until the order form is signed and the data terms are agreed. Greecon keeps the papers; the platform cannot see a signature, so you write down that they exist. In the box above the button write **who signed for the customer and on what date** (for example *Signed for the hotel by Besa Krasniqi on 19 October*): it cannot be left empty. Press **The papers are signed**. Do this at the office, and bring copies on install day.
YOU SHOULD SEE	The step turns **Done**: *Signed and agreed.*, with your name, the time and what you wrote. They are on the handover sheet.
IF YOU DO NOT	Until it is done, commissioning stays grey. The button will not work while the box is empty. If a paper turns out to be wrong, press **Reopen** (if the site had been commissioned, that is taken back too).

Step 2 Prices from the customer's bills

PROJECT MANAGER *optional*

DO	Open **Enter prices and baseline** (once something is entered it is called *Prices and baseline*; the same form is on the site's page). Enter the currency; the electricity price per kWh from the latest bill; what is paid for energy sent to the grid (leave it blank if nothing is paid); the water price per m³; a typical month before (the kWh bought and the m³ of water, from last year's bills); where those numbers came from; whether the site cannot send energy to the grid (tick it if the plant is zero-export); and the time zone, `Europe/Tirane`. Press **Save**.
YOU SHOULD SEE	The step turns **Done** and lists the prices, for example *Electricity: N ALL per kWh* (and *Water: N ALL per m³* where the site has a water flow reading).
IF YOU DO NOT	It stays **To do** and says *Not set: Greecon shows kWh but cannot say what they are worth.* That is allowed: this step is optional and does not hold up commissioning, and without a price Greecon shows kWh only. Without the typical month before, no comparison is shown, which is correct.

Step 3 Install the box

PROJECT MANAGER

DO	Press **Add the box and get its install command** and copy the line it shows (shown once, works once, lasts seven days). Run it on the box exactly as shown. This is done at the office, before install day.
YOU SHOULD SEE	**Done**, with the box's name, its version and *Heard from N seconds ago.* You do not press anything to make it so: the platform saw the box.
IF YOU DO NOT	• *Waiting for the install command to be run on the box.* The line has not been run yet. • *The box has its credential but has not reported yet.* Wait a minute and check the network cable. • *Last heard from N ago: it is offline.* Check the box's power, the network cable and the site's internet. On the box: `greecon-edge status` and `journalctl -u greecon-edge -n 50 --no-pager`. • *Online, but its clock is N minutes fast* (or slow). Run `sudo timedatectl set-ntp true`.

Step 4 Where each device is on the network

PROJECT MANAGER

DO	Press **Find equipment**. Nobody types an IP address, a port or a unit number here. - The box looks at the site's own network (local private addresses only, port 502, about a minute) and only reads. The page says *The box is looking at the network now.* and updates by itself. The platform recognises what answered and fills in at once only what is certain: a device with no address, found at the kit's own number. The rest waits for you. - A device **Found** but not filled in (another address, another number, another meter model): read the line and, if it makes sense, press **Use this address** (or *Use these N addresses*). An optional device the site does not have: press **The site does not have this: remove it**. - The soil probe, the weather sensor and a battery are entered by hand: under the device open **Enter the address by hand** (soil probe 6, weather sensor 7). - Box not on your network (a desk rehearsal), or a converter on another port: open **I know the converter's address**, type its address (and its port, if not 502), and press **Look at this address**.
YOU SHOULD SEE	Each device gets a line in plain words: **Set** (found, address filled in), **Found**, **Not found**, **Not sure which** or **Enter by hand**. When it worked the step is **Done** (*Every device has an address.*), and the button reads **Look again**.
IF YOU DO NOT	- **Not found**: *Nothing that looks like an Eastron meter answers at unit 2. Check the address set on it, the speed, and the A and B wires.* An optional device the site does not have: press **The site does not have this: remove it**. - **Not sure which**: Greecon will not guess, because a wrong guess could swap the grid meter and the solar meter. Read each device's label, use **Enter the address by hand** under each with the number on its label, then **Look again**. Numbers are set at the office: a part with a wrong number goes back there. - *No Modbus converter answered on this network:* check that the converter is on and plugged into the same router as the box. *A converter answers at 192.168.1.50, but nothing answers behind it:* check each meter's address, the speed and the A and B wires. *All answered with exactly the same words:* the converter is not in gateway mode. *The box did not pick this up:* check that it shows as online in the step above. - Readings that stop after it had worked: the router may have given the converter a new address. Press **Find equipment**, then **Use this address**.

Step 5 Which register holds which reading

PROJECT MANAGER

DO	The Eastron meters and the pump contact module need nothing typed. For the 4 to 20 mA module the page asks, for each transmitter, only for the range printed on its label, in the label's own words: - **Level transmitter** (input 1): the **Range on the label**, for example 5, its **Unit on the label** (metres, or bar) and **How tall the tank is inside** in metres, for example 4: from the bottom to the top of the water when full. - **Pressure transmitter** (input 2): the **Top of the range** in bar; 10 is usual. - **Flow meter** (input 3): the **Top of the range** in litres a minute. Greecon works out the register and the scale. For a transmitter the site does not have (no flow meter), tick **Not fitted at this site**: Greecon asks first, then takes the reading and its history away. Press **Save the ranges**. Equipment Greecon does not know (the soil probe, the weather sensor, a battery) is mapped by hand, from the maker's register table: under the reading press **Choose the register** and fill in the register kind (**Input** for a manual's 3xxxx, **Holding** for 4xxxx), the address (counted from zero: 30053 is address 52, 40001 is address 0), the kind of number (and which half comes first), how it becomes the reading (**Multiply and add**, or **A straight line, such as 4-20 mA**) and which part of the value. Press **Save the register**.
YOU SHOULD SEE	**Done**: *Every reading is tied to a register.* Under each transmitter the form says what it set, for example *Set now: 4 mA is 0 % and 20 mA is 125 %.* (0 to 5 m in a tank 4 m tall); the tank level is a percentage of the tank, so a full tank reads 100.
IF YOU DO NOT	*N readings not tied to a register yet*, each marked *Needs the range printed on the transmitter's label.* (the 4 to 20 mA module) or *No register chosen.* (equipment Greecon does not know). A transmitter shorter than the tank is refused: *The transmitter reads up to 3 m and the tank is 4 m tall: it cannot see the top of the water. Fit a transmitter with a longer range.* A reading the site does not have holds the list here: tick **Not fitted at this site**, or press **The site does not have this: remove it** (step 4) for an optional device.

Step 6 Every device answers the box

THE PLATFORM

DO	Nothing to press: the platform does this one by itself once the box is online and each device has its address and registers. Wait a minute.
YOU SHOULD SEE	Each device shows **Done**: *Answering (N readings)*. On the box, `greecon-edge status` lists each device as **ok**.
IF YOU DO NOT	The device shows **Needs attention** with the reason in words, for example *no answer from 192.168.1.50:502 (unit 2)*. Look in this order: the converter's address may have changed (press **Find equipment** again, then **Use this address**); the number on the device's label against the install card; the speed at the converter; A and B swapped on the data cable; the converter not in gateway mode. Nothing answers at all: the converter is off, or it is on another network than the box. Test one register from the box with `greecon-edge modbus-test` (see the box commands): if it prints *Could not read*, the problem is the wiring or the settings, not Greecon.

Step 7 Readings arrive and look right

THE PLATFORM

DO	Nothing to press. Look at each reading: is it a number you believe?
YOU SHOULD SEE	Each reading shows **Done** with its value and age, for example *N %, 4 seconds ago*, and the step says *Every reading is arriving and could be real.* **Pump running** reads 1 while the pump runs and 0 while it is stopped.
IF YOU DO NOT	*No reading yet*: wait a minute. *Last reading N ago. It has stopped arriving*: a device or the converter lost power, or a cable came loose. A value that cannot be real says so, for example *-31.2 cannot be a tank level (0 to 100). Check the register address, the scale or the sensor.* On a transmitter a reading below zero is nearly always a wire that has come off (the module then reads 0, below the 4 mA of an empty tank); one far too high is a range or tank height typed wrongly: give the ranges again (step 5).

Step 8 Someone is told when the site needs attention

PROJECT MANAGER

DO	Open **Who is told**. **Add a person** for each name from the survey, in the order they should be told (each needs an email, a mobile number with its country code such as `+355 69 123 4567`, or both; give everyone an email address as well as a number). Press **Save the list**. Press **Send a test message** beside each person and ask them whether it arrived: in the inbox, not in spam; on the phone. Tell each person that every alert message has an *I have seen this* link: one tap, no sign-in, and it stops the message going on to the next person. Give a login of their own (the Operator role) only to those who should also look at the screens. Then, back on the setup list, press **I checked this**.
YOU SHOULD SEE	After the test: *A test message was sent N ago. Greecon cannot tell whether it was read.* After you press the button: **Done**, *A test message reached the person on the list.* The button stays grey until the list has someone on it and a test has been sent.
IF YOU DO NOT	▪ *This Greecon platform is not set up to send messages:* tell the founder; the email service must be switched on before a site can be commissioned. ▪ *Nobody is set to be told:* add the people. ▪ *Nobody on the list has an email address* (or a phone number): add one for each person. ▪ A message that did not arrive: look at **What was sent** under **Who is told**, check the address and the country code, and ask about spam. ▪ *Test messages are limited to 6 an hour.* Wait and try again.

Step 10 Alert limits agreed with the customer

PROJECT MANAGER

DO	The step lists each reading that has limits, with what applies. Go through them with the customer and agree numbers they would act on. To change one, open the **Standard** drop-down beside the reading (it says *1 set*, *2 set* once changed), type the numbers and press **Save limits**. When they are agreed, press **I checked this**.
YOU SHOULD SEE	Each reading with limits is listed with its numbers, for example *Tank level: Critical below 15 % (standard); Low below 35 % (standard)*. After you confirm: **Done**, *Checked on site.*
IF YOU DO NOT	The standard limits apply until you change them: tank **Critical below 15 percent** (an Act now alert) and **Low below 35 percent** (a Check today alert); pump **No flow at or below 0.2 litres a minute** (an Act now alert once it has held for 30 seconds, raised only when the box reads both a flow and the pump's running state); battery **Low below 25 percent**. **Start irrigating below** has no standard value: Greecon gives no irrigation advice until the grower's own value is entered.

Step 11 Electrical work done by a licensed electrician

PROJECT MANAGER ELECTRICIAN

DO	Fitting meters in a board and wiring transmitters and contacts is the **licensed electrician's** work, under the customer's lock-out. When the electrician has finished and the lock-out is removed, ask for their name and licence number, photograph the wiring, the labels and the displays, and check that the lock-out has been removed. Then write in the box the electrician's **name and licence number** (for example *Arben Hoxha, licence E-1234*), or that no electrical work was needed: it cannot be left empty. Press **The electrical work is done**.
YOU SHOULD SEE	The step turns **Done**: *Done by a licensed electrician.*, with your name, the time and what you wrote (they are on the handover sheet). The platform does not check the licence: what you write is the record.
IF YOU DO NOT	Until it is done, commissioning stays grey. Do not press the button while the work is open or the lock-out is still on. The button will not work while the box is empty.

Step 12 Commissioned

PROJECT MANAGER

DO	Greyed out until every other step is done (optional steps do not hold it). Then it says: *On site, compare what Greecon shows with what you can see: the meter's own display, the tank, the pressure gauge. When they agree, confirm here.* Do the commissioning tests in this pack, write the pairs of numbers on the record, then press **Commission this site** with a note.
YOU SHOULD SEE	The progress line reads **Commissioned.**, your name and the date are on the handover sheet, and **Open the handover sheet** appears.
IF YOU DO NOT	While it is grey it says *N steps above must be done first.* and *Finish the steps above.* Finish them first.

Commissioning

Commissioning is checking, once, on site, that what Greecon shows is what the real world is. PROJECT MANAGER with the electrician and the owner's own person.

- Compare in the order of the table, and write the pair of numbers (what Greecon shows, what you can see) on the record.
- **Do not start, stop or open any equipment yourself.** The owner's own person does, because they know what else it affects.
- A check that fails is not commissioned: find the cause (the three most likely problems are at the end of this pack), fix it and check again. When every line passes, press **Commission this site** on the setup list (step 12) with a note. It stays grey until every other step is done, the papers (step 1) and the electrician's name and licence number (step 11) included. The progress line then reads **Commissioned.**, and your name and the date are on the handover sheet.

DO

The internet cut, step by step. Cut only the site's internet, for ten minutes: unplug the router's uplink cable (the one that goes to the wall or the modem), or take the SIM out of the cellular router. **Leave the box and the converter plugged into the router:** if the box cannot reach the converter it reads nothing, and there is nothing to test. While it is out, `greecon-edge status` shows a queue. Then restore the internet. A few minutes after it returns the queue is empty and the readings from the gap are in the downloads. **Write the result down:** it is the evidence for the statement that the box keeps its readings.

Check and how	Pass when	Greecon shows	Real	Result
Soil probe. In the air, then in a glass of water, then in the soil.	Three different readings: low in the air, high in the water, plausible in the soil			☐ Pass ☐ Fail
Weather sensor. Against a hand-held thermometer at the sensor. (This check is added by the pack: the launch documents give no tolerance, so write both numbers and judge.)	Plausible for the time of day			☐ Pass ☐ Fail
Tank level. A dipstick or the sight gauge, at two levels.	Within 5 percentage points			☐ Pass ☐ Fail
Pressure. Against the pressure gauge on the same line.	Within 0.2 bar			☐ Pass ☐ Fail
Flow (if there is a flow meter). A bucket test (fill a measured volume, time it), or the flow meter's own display.	Within 10 percent			☐ Pass ☐ Fail
Pump running (if there is a pump to watch). The owner starts the pump.	Greecon shows running within 30 seconds, and changes both ways			☐ Pass ☐ Fail
4 to 20 mA channels. With a loop calibrator, send 4, 12 and 20 mA.	The reading at the bottom, the middle and the top			☐ Pass ☐ Fail

Check and how	Pass when	Greecon shows	Real	Result
A test message. Each person on the list.	It arrived, in the inbox and on the phone			☐ Pass ☐ Fail
Power cut. Unplug the box for 30 seconds, plug it in.	Back reporting within 3 minutes, and the box says nothing about readings lost			☐ Pass ☐ Fail
Internet cut (the first install, and any time the box or its software is new). See the box above.	A queue while it is out; empty a few minutes after it returns			☐ Pass ☐ Fail

Signed off by: _____ Date: _____

Handover

About 30 minutes at the customer's, with the owner and the owner's people who are told. PROJECT MANAGER

- ☐ **Print the handover sheet.** On the setup page open **Open the handover sheet**, check it, and print it (**Print or save as PDF**; the printed page leaves out the menus). It names the customer and the site, says where each device is (address, unit number, registers), who is told, what was commissioned and by whom, who signed the papers and who the electrician was, and how to reach Greecon (`info@greecon.earth`).
- ☐ **Show the owner, in ten minutes, in this order:** the **Overview** (is everything all right, in one sentence); the site's own page under **Sites**, with the soil moisture, the air temperature and humidity, the tank level and the pressure; **Who is told** (under **Settings**) and a live test message arriving on their phone, and that each message carries an *I have seen this* link, so that nobody needs a login to acknowledge an alert; **Reports**, under **Downloads**.
- ☐ **Write the probe's depth and place** on the handover record, and say that soil readings are only as good as the contact between the probe and the soil.
- ☐ **Say that Greecon gives no irrigation advice yet,** and agree when you will ask what reading means "time to water" (after the first week).
- ☐ **Ask at what air temperature frost damages the crop, and in which weeks.** Enter it as the frost limit; say that the alert is an Act now message, possibly in the night, and agree who is woken.
- ☐ **Say the two things in the boxes below, in these words:** what Greecon is, and how an alert reaches people.
- ☐ **Leave** the printed sheet, signed; the box's label; and a card with `info@greecon.earth` and your phone. **File** the photographs and the survey form in the shared drive, in a folder named for the customer.
- ☐ **Send the third message on The install card** to the customer's contact the same day, with the handover sheet attached (its Albanian is a draft: have it read first).
- ☐ **Leave the customer's account:** press **Back to the Greecon console** (the banner's button). It returns you to **Today**.

SAY

Greecon watches your equipment and tells you. The protection built into your equipment remains the safety layer. You will get one message each morning. If there is nothing to do, it says so.

SAY

How alerts reach people: **Act now** *goes by email at once and by text ten minutes later if nobody has acknowledged it; thirty minutes after the alert began, the next person is told by email and text, and so on down the list. Saying you have seen it means pressing the I have seen this link in the message (no sign-in, good for 72 hours) or* **Acknowledge** *on the* **Alerts** *page (for a person with a login).* **Check today** *is one email, held until morning if it is night. Beyond those, three short notes only: one to the people who were told when somebody acknowledges the alert (who took it), one when the problem has stayed clear for 15 minutes (***Resolved***), and one morning message at 07:30 saying whether everything is all right. Quiet hours (22:00 to 07:00) hold everything except Act now.*

STOP

Never promise a saving, a percentage or "AI". Say what Greecon measures and shows. The product shows what the customer's own solar kept on site, at the customer's own prices, from real readings. That is the claim. And say that Greecon is not a safety system.

Their data is theirs. Every reading for the last 31 days, hourly averages for up to a year, the alerts, the audit log and how the account is set up can be downloaded whenever they want, from **Reports** under **Downloads**. If they stop, they can ask Greecon for every reading it holds, as a file.

Handover record

KIT F · FARM OR GREENHOUSE WITH IRRIGATION

Field	Answer
Customer and site	
Date, and who was there	
Who is told, in order (name, mobile with country code, email)	
Papers: who signed for the customer, and the date	
Electrician: name and licence number (or: no electrical work was needed)	
Each probe: depth, and where it is	
Alert limits agreed (tank Critical below, Low below; no-flow level if a pump is watched)	
“Start irrigating below”: not set yet; to be asked after the first week, on	
Setup list: every step Done and the site Commissioned	☐ Yes
Handover sheet printed and signed	☐ Yes
Shown: Overview, the site’s page, Who is told with a live test message, Reports	☐ Yes
Left: label on the box; card with `info@greecon.earth` and your phone	☐ Yes
Photographs and survey form filed in the customer’s folder	☐ Yes

Project manager: name and signature

Owner: name and signature

The three most likely problems

What you see	Likely cause	What to do
A probe or the weather sensor does not answer	A wrong unit number (many are delivered as unit 1, the same as another device: that is why every number is set at the office, before the probe is buried), a number or an address entered by hand wrongly, a speed or parity that does not match the converter, A and B swapped, no power	Check each in that order. Test the register from the box (`greecon-edge modbus-test`) against the maker's table. A device whose number was set before burying is easy to check; one that was not must be dug up, so every number is set before the probe goes in the ground
The soil reading looks wrong (constant, or the same in air and in water)	A wrong register, the wrong register kind (Input or Holding), a wrong scale, or poor contact between the probe and the soil	Re-test the register against the maker's table; repeat the air, water and soil check; check the probe's depth and contact
A frosty morning reads as a hot afternoon	The air temperature was mapped as an unsigned number	Map it as int16 (signed), with the scale of the sensor's table
No irrigation advice is shown	"Start irrigating below" has no standard value	After the first week of readings, ask the grower what reading means "time to water" and enter it as the limit (setup step 10)

More, by symptom, are in the Appendices of the Complete Deployment Guide (Appendix C) and in the Field Manual. A device that does not answer is, in this order: the converter's address changed (press **Find equipment** again), the number on the device's label not the one on the install card, a wrong speed at the converter, A and B swapped on the data cable, the converter not in gateway mode.